Collision Course:

Exploring the effects of mergers on dark matter halo dynamics

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PHYS 410
Fall 2025

What is dark matter?

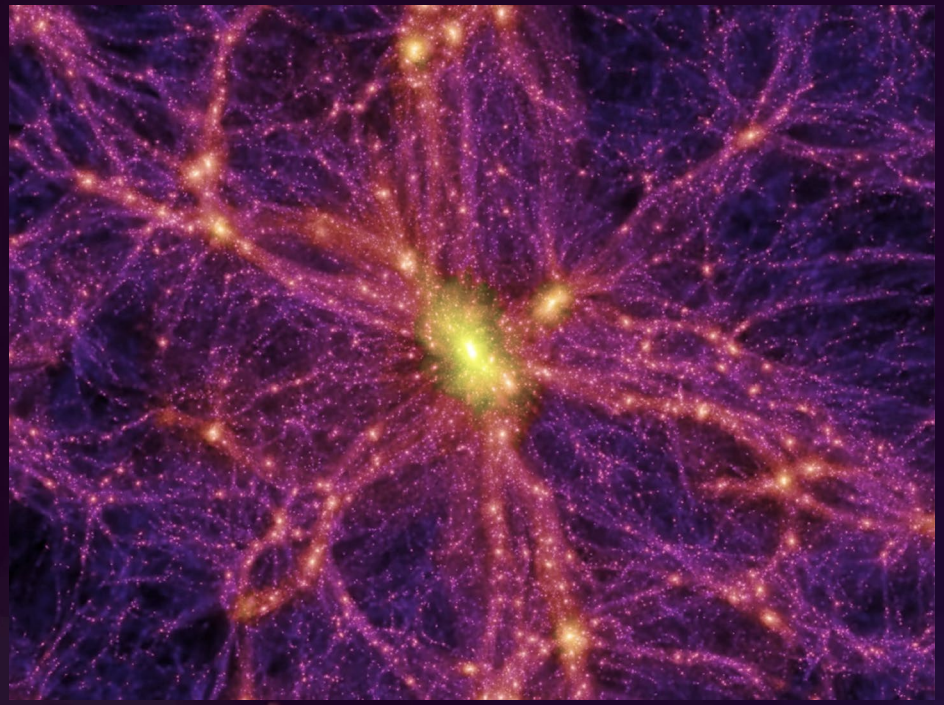


Image: Springel Volker from National Georgraphic

Missing mass = ‘cold’ dark matter

Enter: the ΛCDM model (Tested by Klypin et al. 2011)

Dark matter grows hierarchically

Dark matter halos

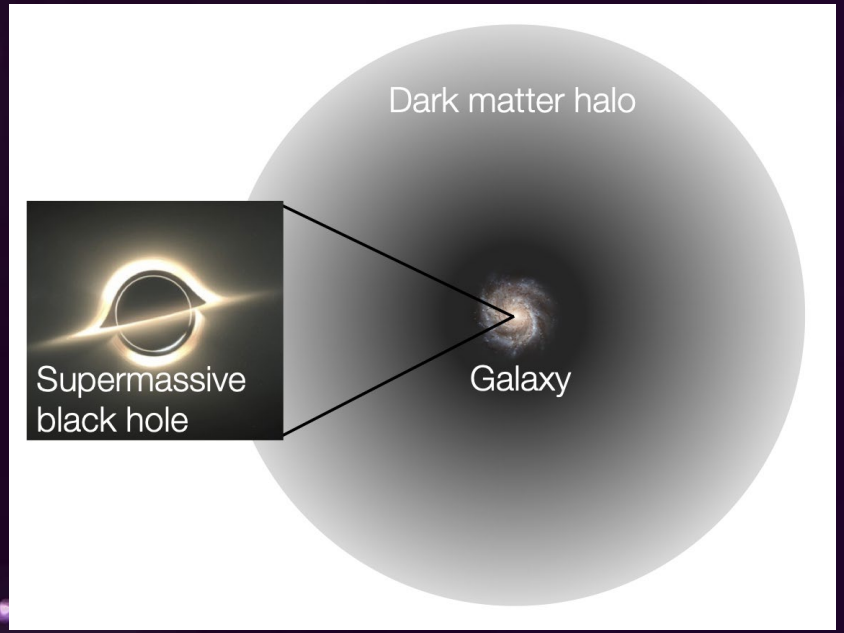


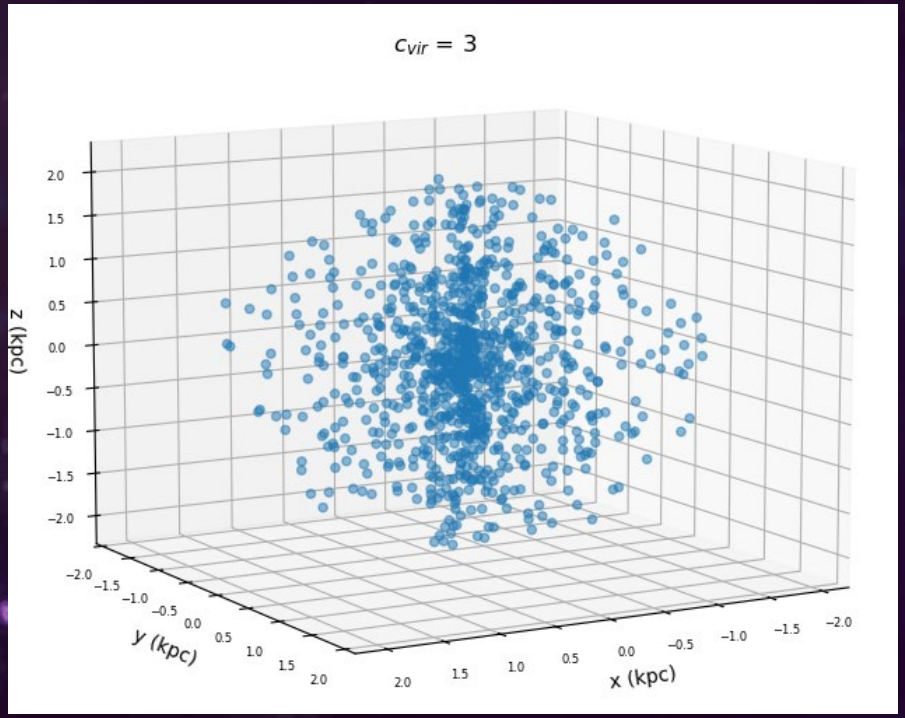
Image: Powell, M. From Stanford.edu

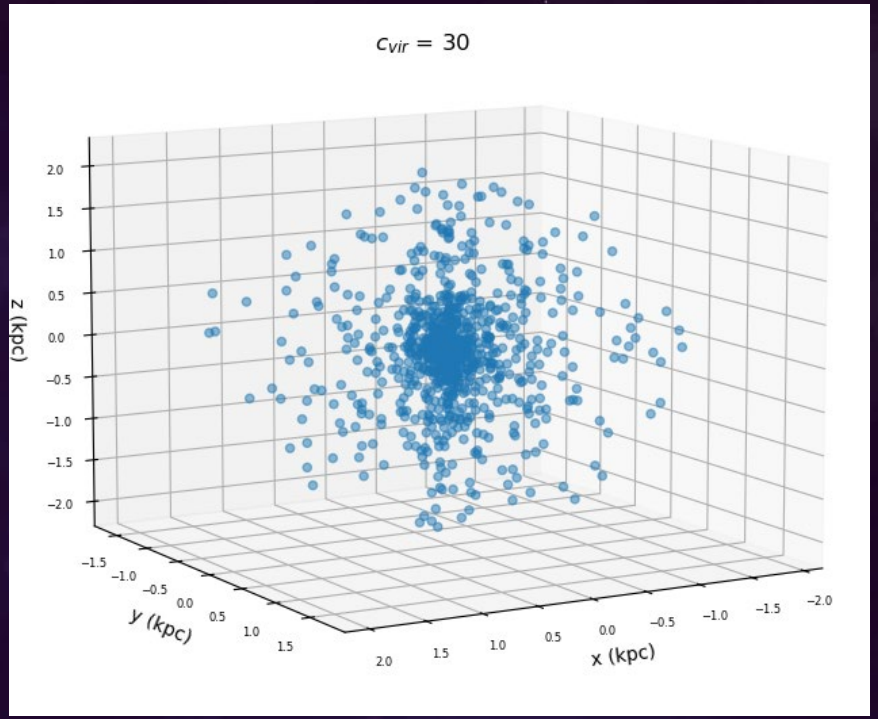
Potential wells collect gas + dust

Galaxies are born in densest inner region

Concentration

A measure of how centrally dense a dark matter halo is





Changes to internal density: Mergers

Dynamical friction

The satellite halo pulls host particles to higher velocity

Thus increasing the kinetic energy and decreasing the density

Tidal shocks

Highly uneven potential field with rapid changes compared to particle orbits

Particles move rapidly between halos decreasing the central density of both

Evidence of disturbances: Wang et al. 2020

Sampled halo groups from high res FIRE-2 simulation

Found that both types affect the inner density

But their focus was changes from redshift

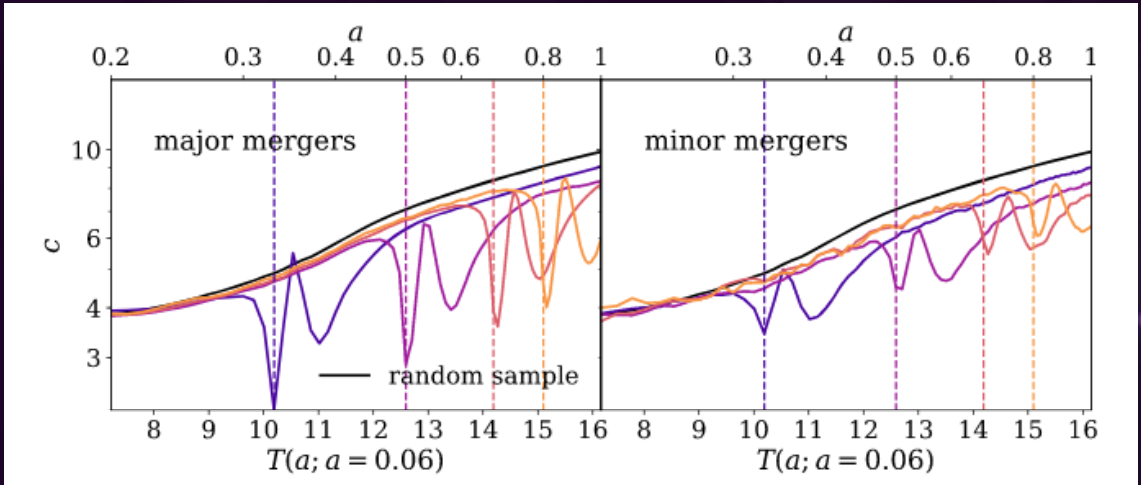


Image: Wang et al. 2020

The central question:

What is the relation between the mass ratio of a merge and the change in concentration of the host halo?

Strategy

- Simulate a controlled* collision between two dark matter halos using N-body code

- Not included: baryonic matter or dark energy -> only gravitational forces apply

$$\vec{a}_i = \sum_{i \neq j}^{N} \frac{G m_i \overrightarrow{r_{ij}}}{\left|\overrightarrow{r_{ij}}\right|^3}$$

$\vec{a}_i$ = particle acceleration
m_i = particle mass
r_{ij} = distance between particles

Strategy

- Evolve until virialization of the halo is stabilized

$$T = \frac{U}{2}$$

- Measure the concentration of the halo through time and compare each merger type

$$c_{vir} = \frac{r_{vir}}{r_{scale}}$$

Measuring concentration

- The NFW profile defines the density of dark matter only halos.

$$\rho(r) = \frac{\rho_{\text{scale}}}{(\frac{r}{r_{\text{scale}}})(\frac{r}{r_{\text{scale}}} + 1)^2}$$

- rscale defines where the density changes rapidly

$$Q^2 = \frac{1}{N_{\text{bins}}} \sum_{i}^{N\text{bins}} [log_{10}\rho_i - log_{10}\rho_i^{model}]^2$$

- Multiple fits performed + best fit determined by Qmin

The simulation aides

1 **Leapfrog method**

Uses half-step method to obtain a second order accuracy with one acceleration call

2 **Barnes and Hut tree**

Reduces force computations by dividing the space into a framework that approximates forces from far away sectors

3 **Scipy.optimize's curve_fit**

The density profile of the halo will be fit to the NFW equation using bounds for both parameters – varying rscale bounds

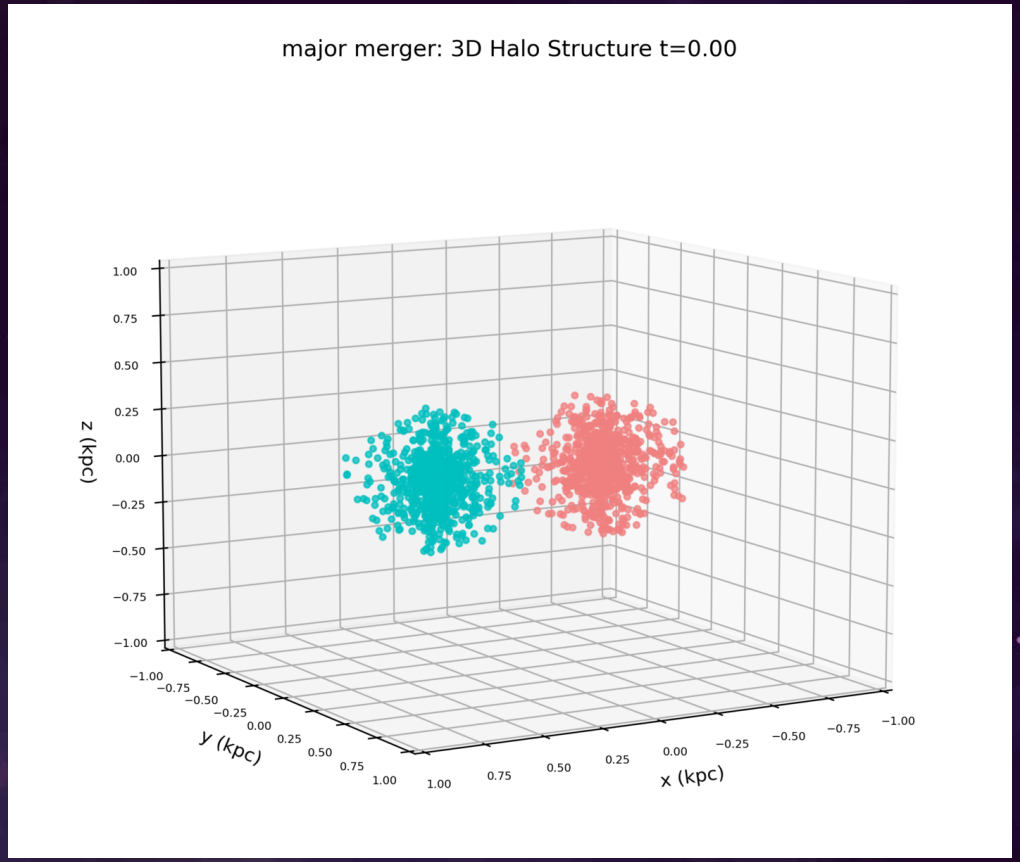
Mass ratio: μ

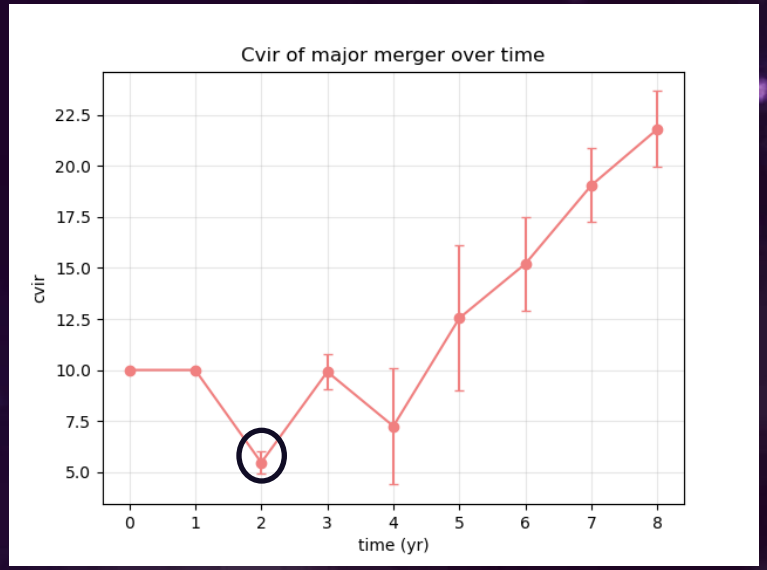
I ran the simulation for 3 mass ratios between the sizes of major and minor mergers:

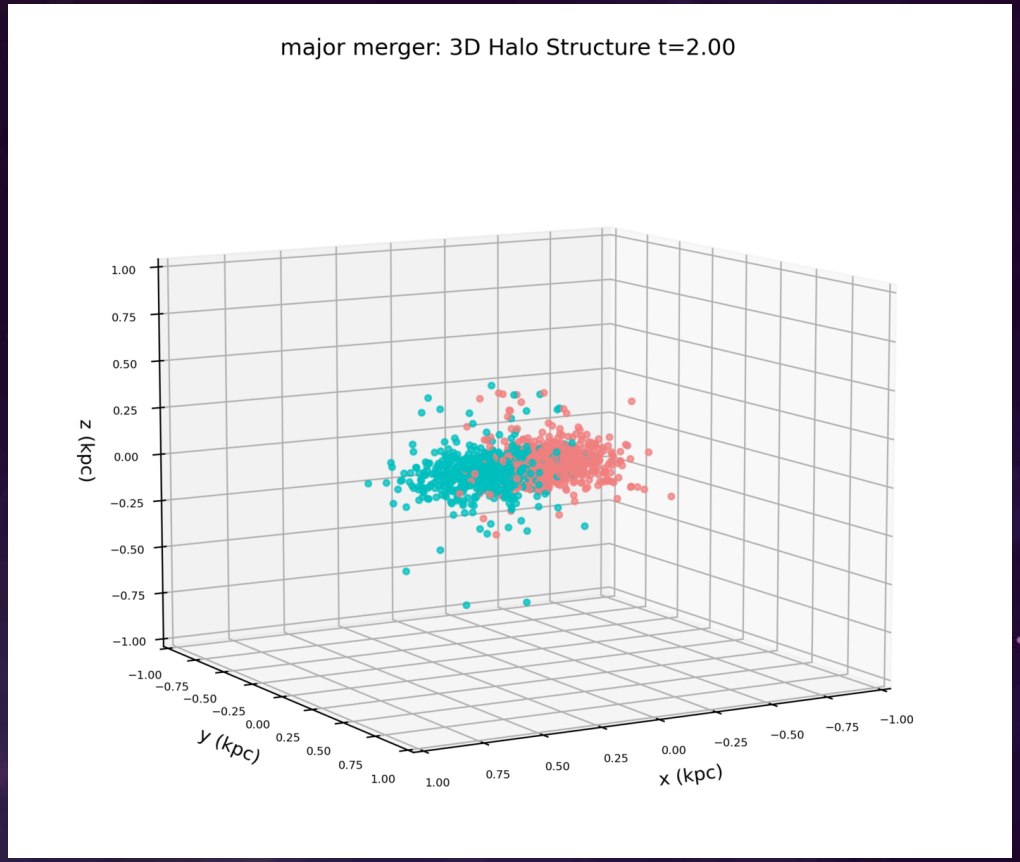
$$\mu = \frac{M_{host}}{M_{satellite}} = [1, 5, 10]$$

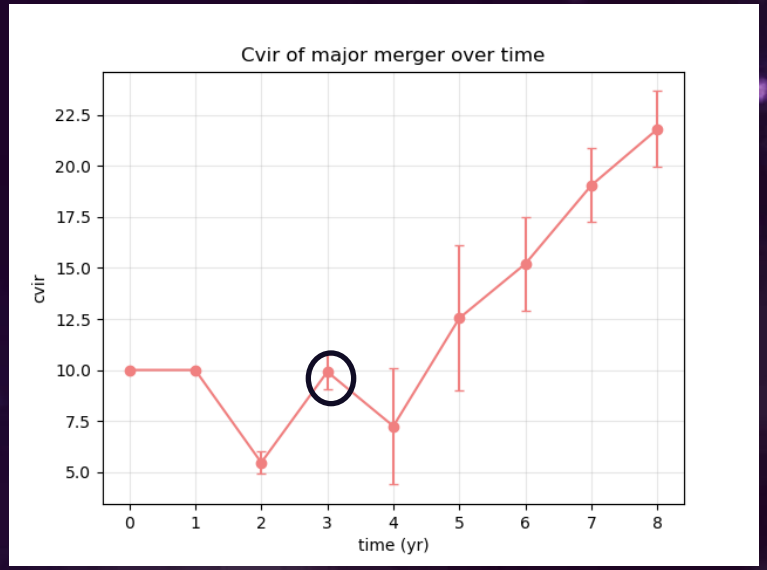
Particles have equal mass so satellite will remain $M_{min} = 10^3 M_{sun}$

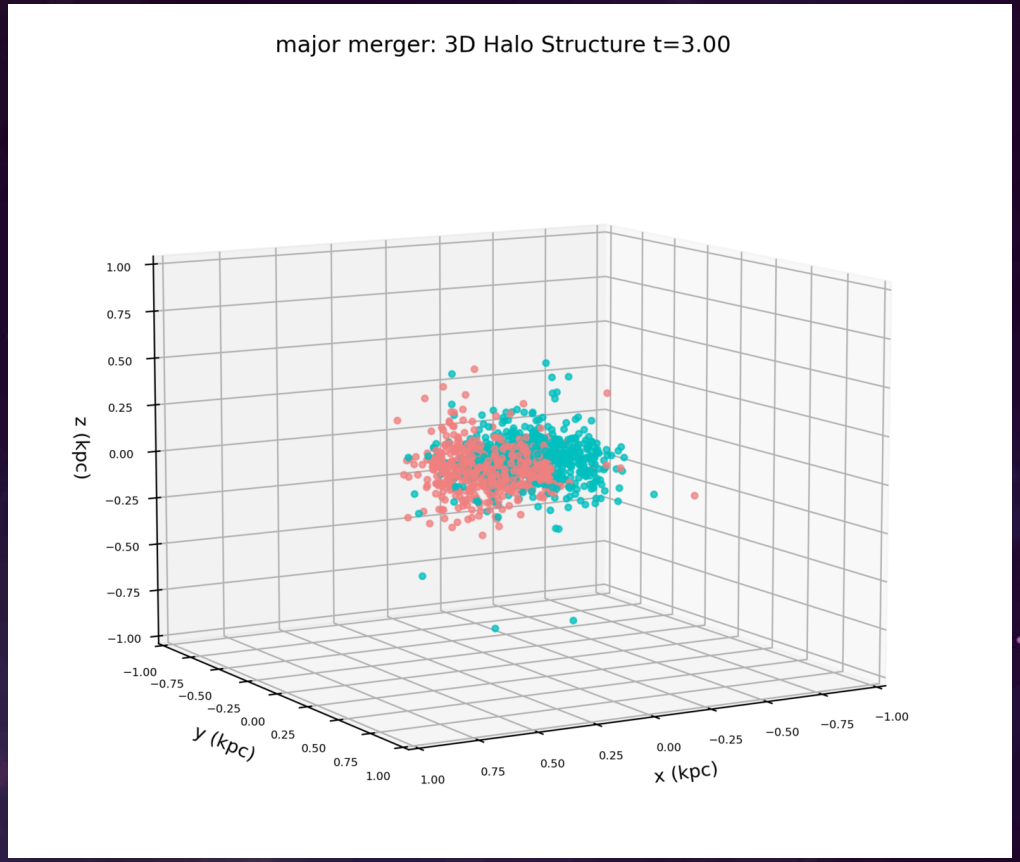
$$M_{host} = [10^3, 5 \times 10^3, 10^4]\ M_{sun}$$

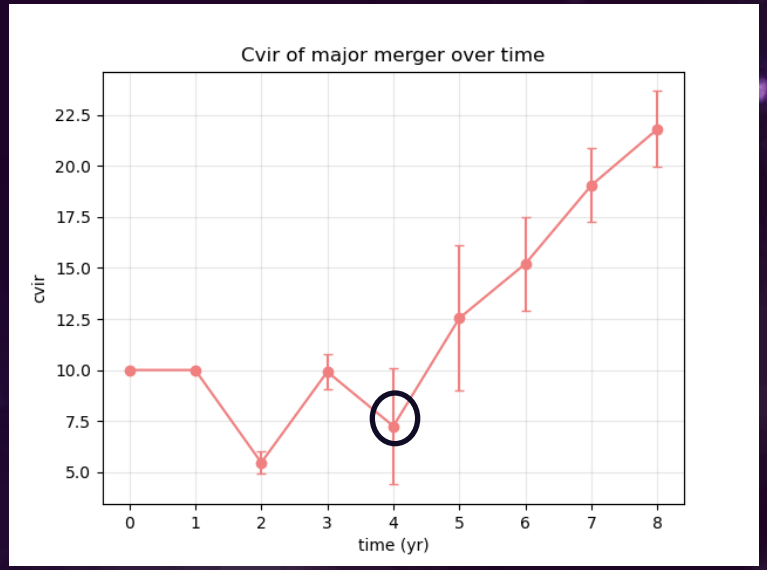

major merger: 3D Halo Structure t=0.00
z (kpc)
y (kpc)
x (kpc)

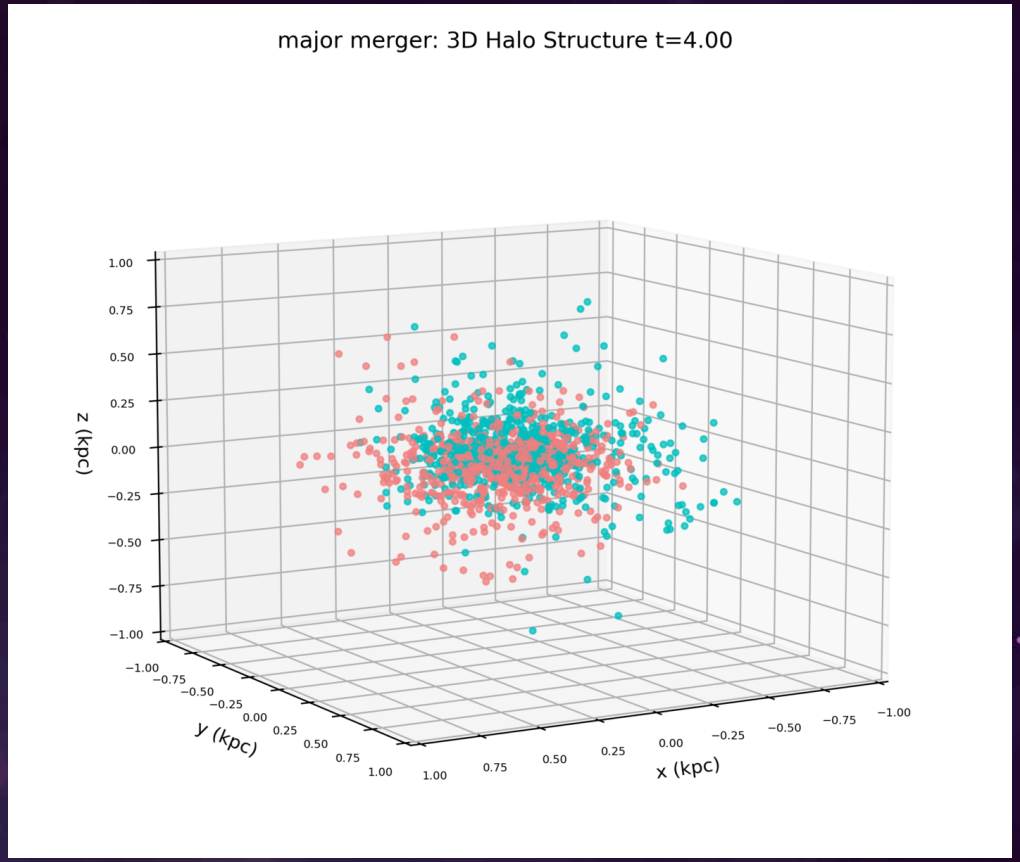

Cvir of major merger over time
cvir
time (yr)

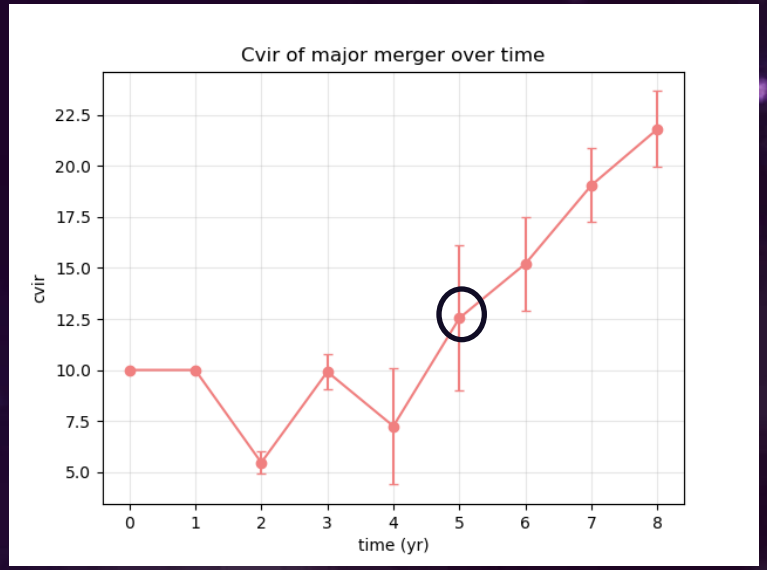

major merger: 3D Halo Structure t=2.00
z (kpc)
y (kpc)
x (kpc)

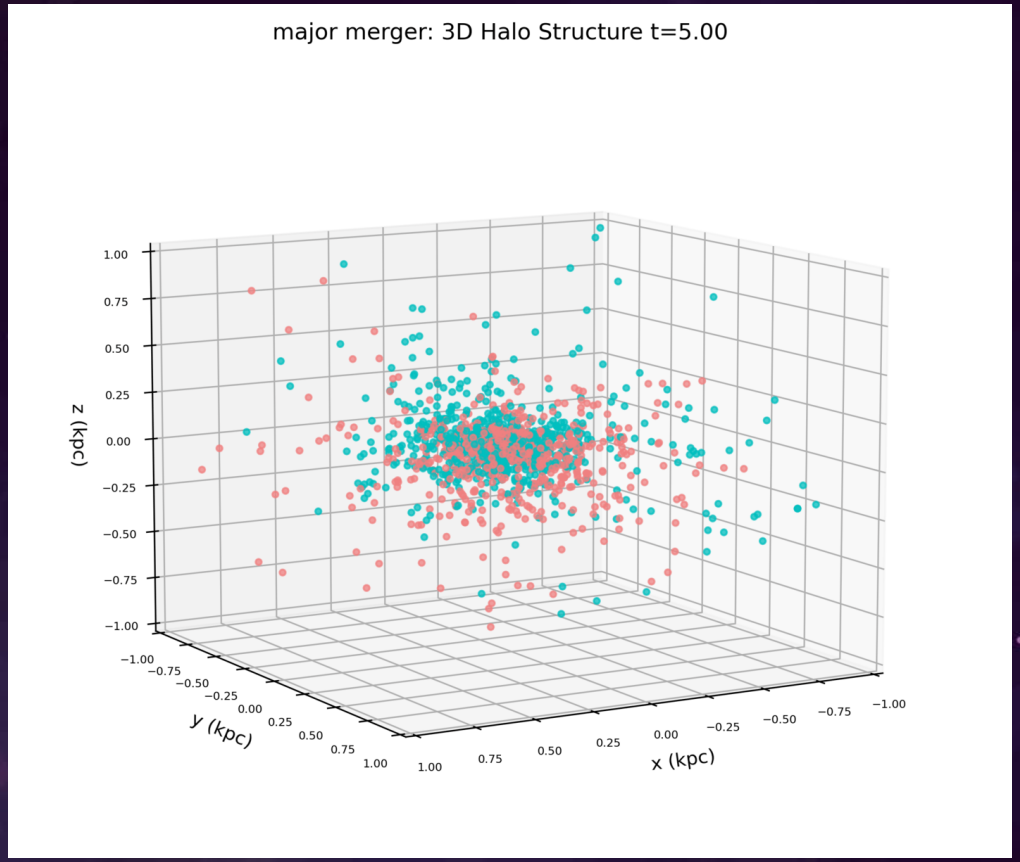

Cvir of major merger over time
cvir
time (yr)

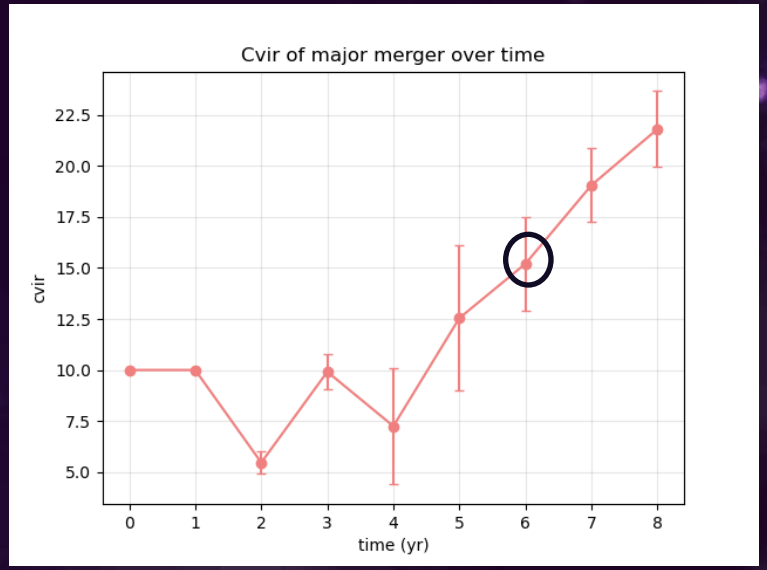

major merger: 3D Halo Structure t=3.00
z (kpc)
y (kpc)
x (kpc)

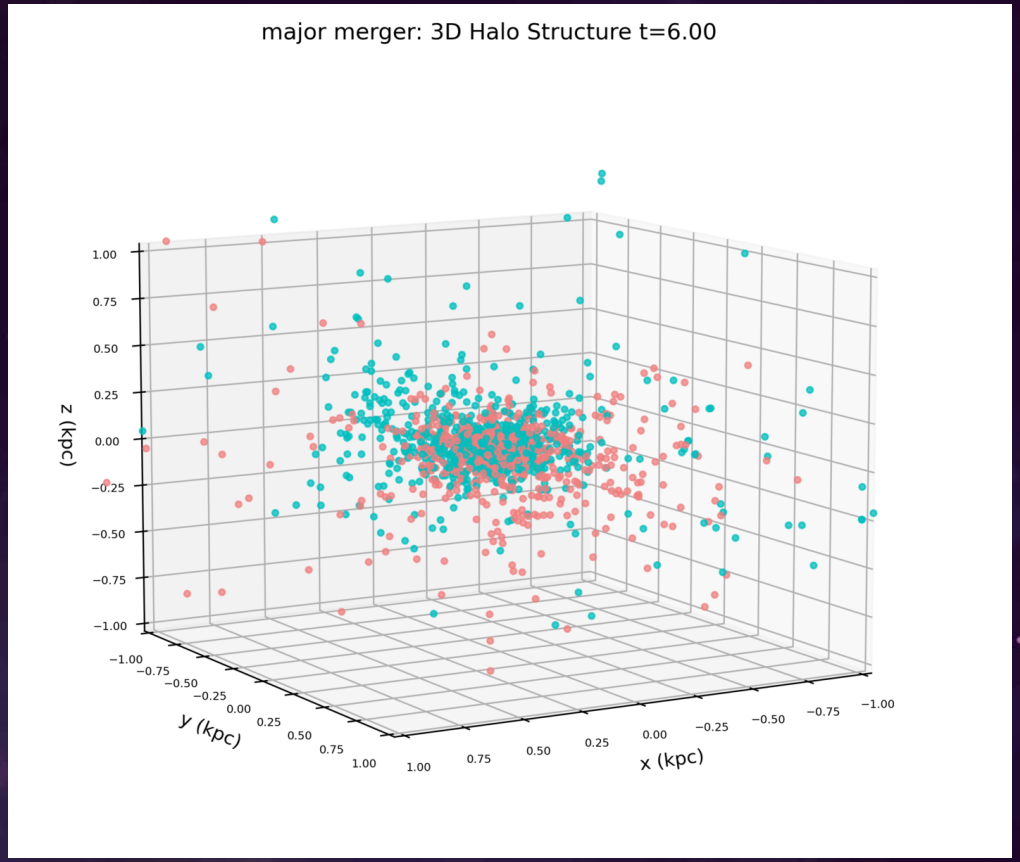

Cvir of major merger over time
cvir
time (yr)


major merger: 3D Halo Structure t=4.00
z (kpc)
y (kpc)
x (kpc)

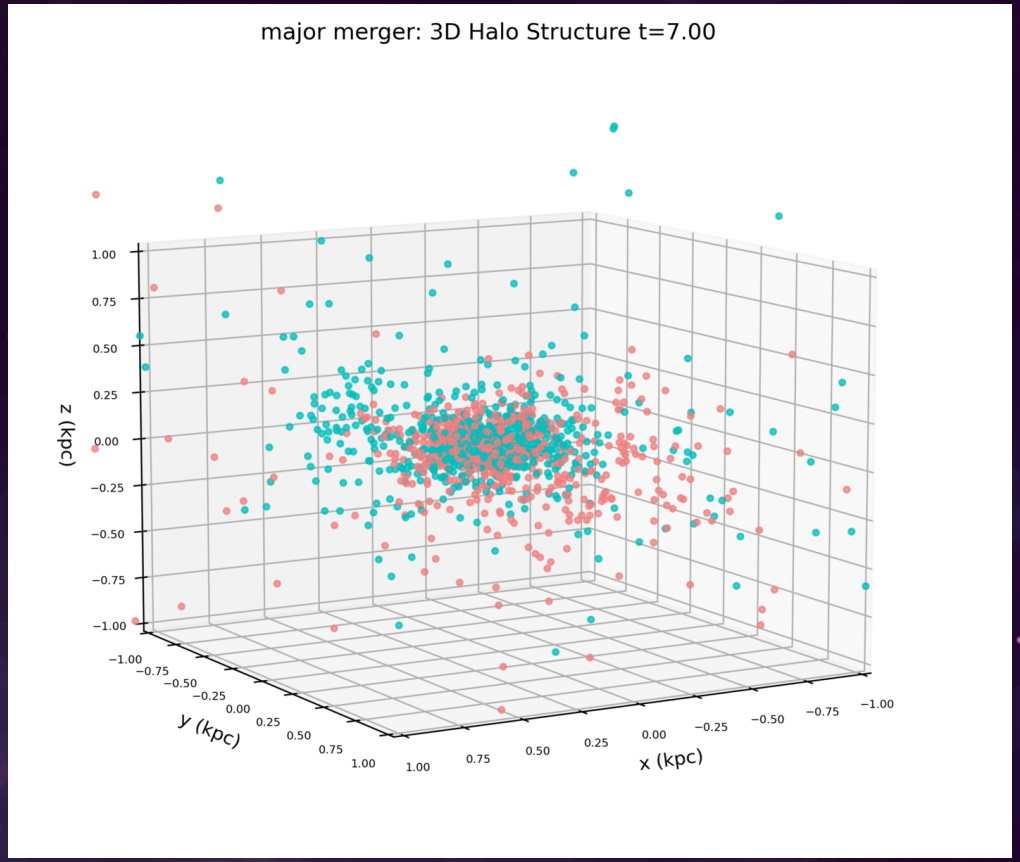

Cvir of major merger over time
cvir
time (yr)


major merger: 3D Halo Structure t=5.00
z (kpc)
y (kpc)
x (kpc)

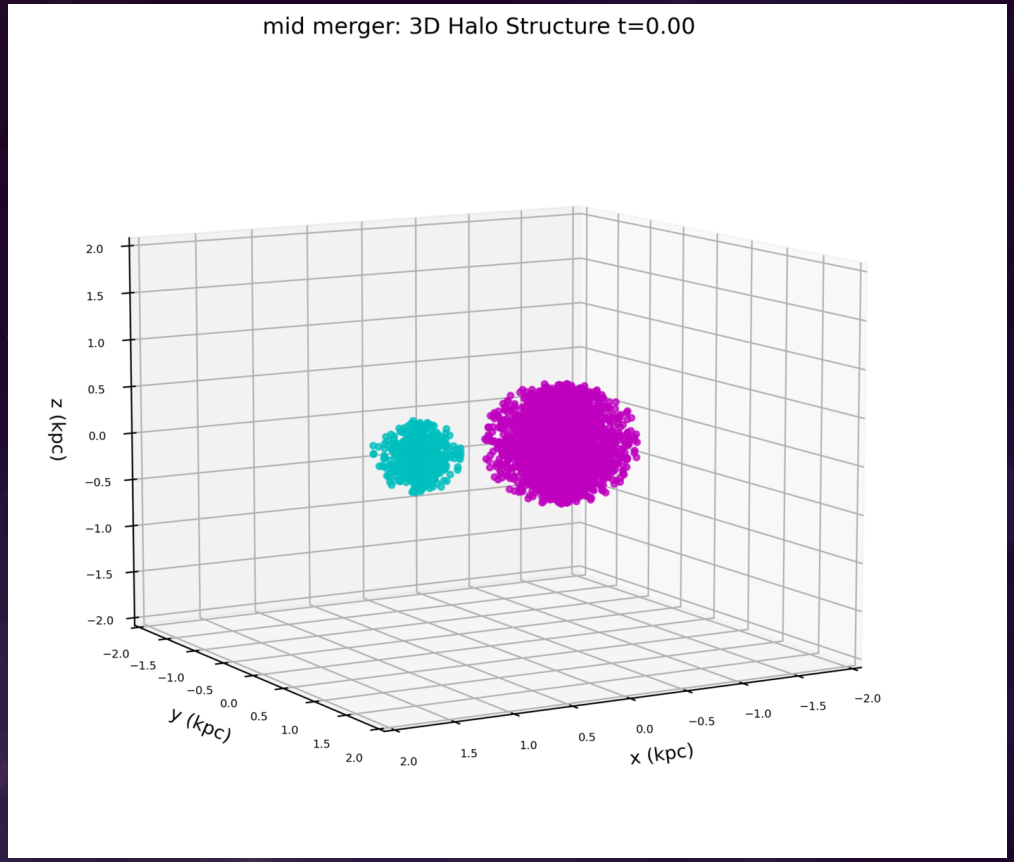

Cvir of major merger over time
cvir
time (yr)

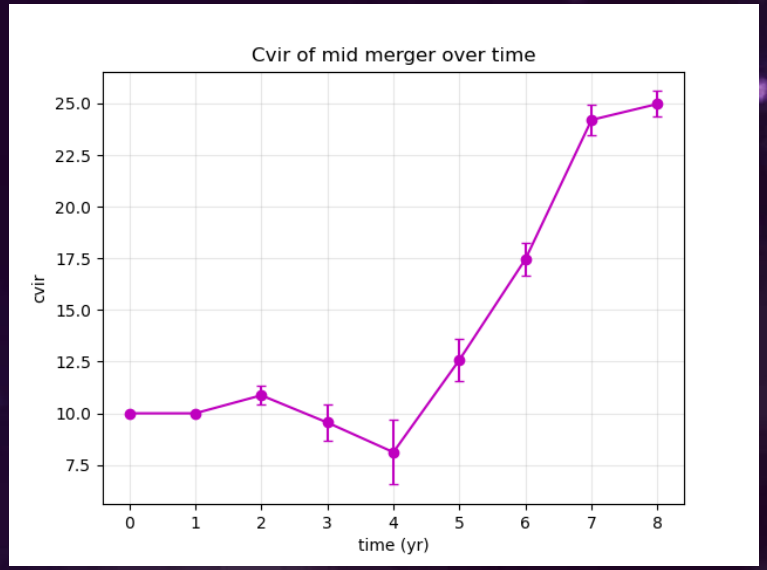

major merger: 3D Halo Structure t=6.00
z (kpc)
y (kpc)
x (kpc)
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

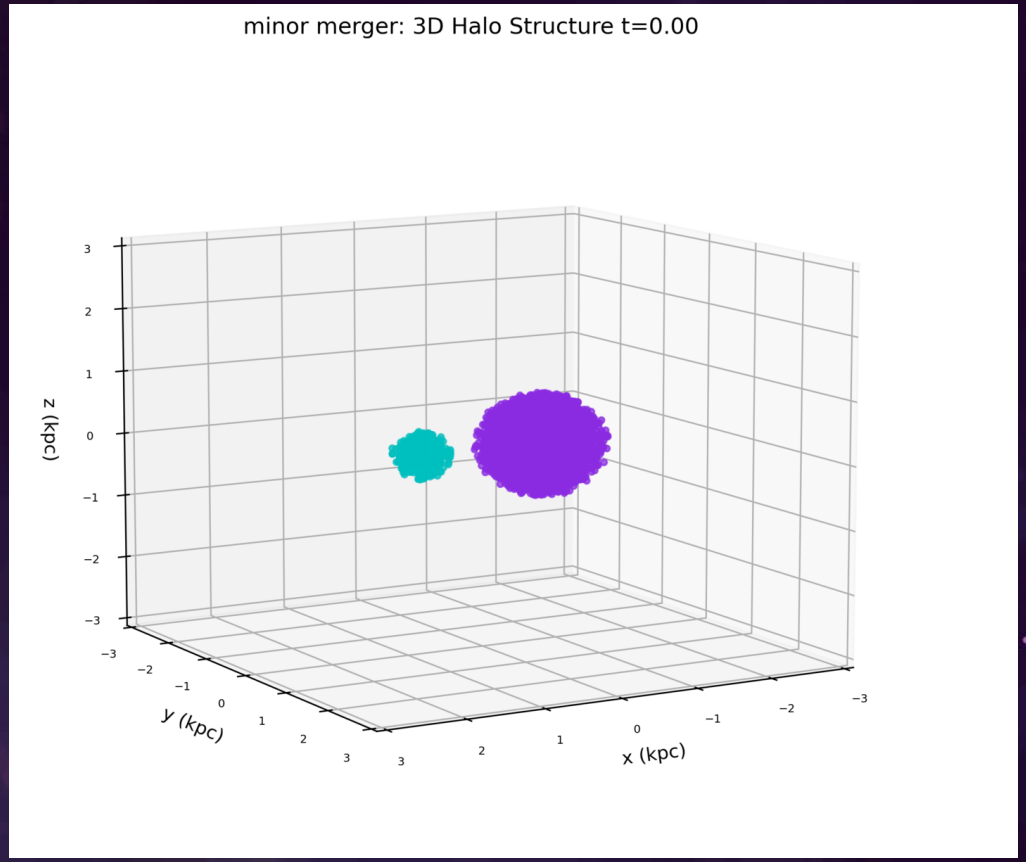
Cvir of major merger over time
cvir
time (yr)
22.5
20.0
17.5
15.0
12.5
10.0
7.5
5.0
0
1
2
3
4
5
6
7
8


major merger: 3D Halo Structure t=7.00
z (kpc)
y (kpc)
x (kpc)
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

Cvir of major merger over time
cvir
time (yr)
22.5
20.0
17.5
15.0
12.5
10.0
7.5
5.0
0
1
2
3
4
5
6
7
8

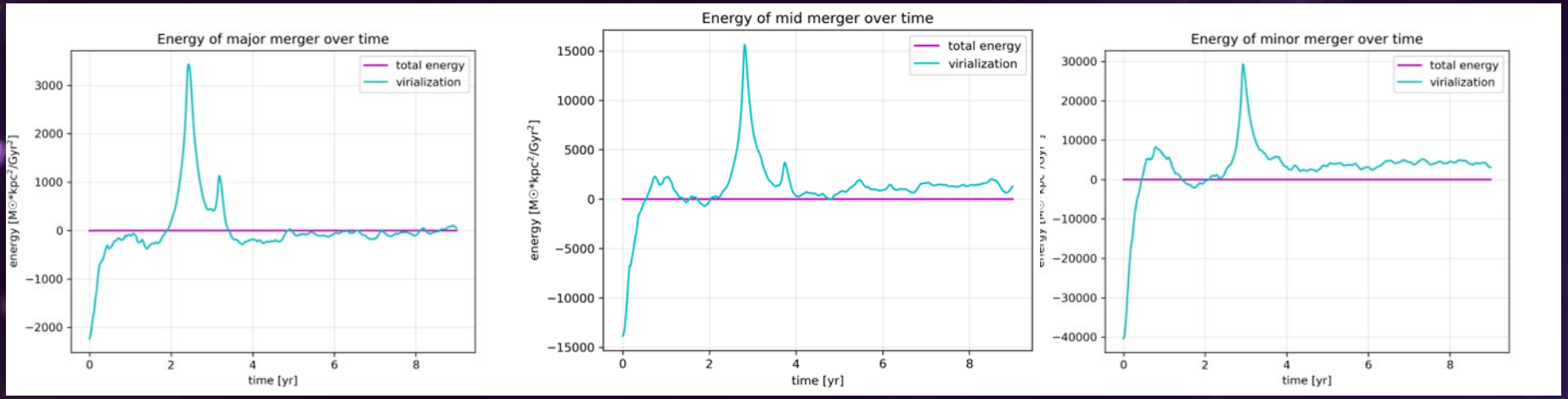

mid merger: 3D Halo Structure t=0.00
z (kpc)
y (kpc)
x (kpc)


Cvir of mid merger over time
cvir
time (yr)


minor merger: 3D Halo Structure t=0.00
z (kpc)
y (kpc)
x (kpc)

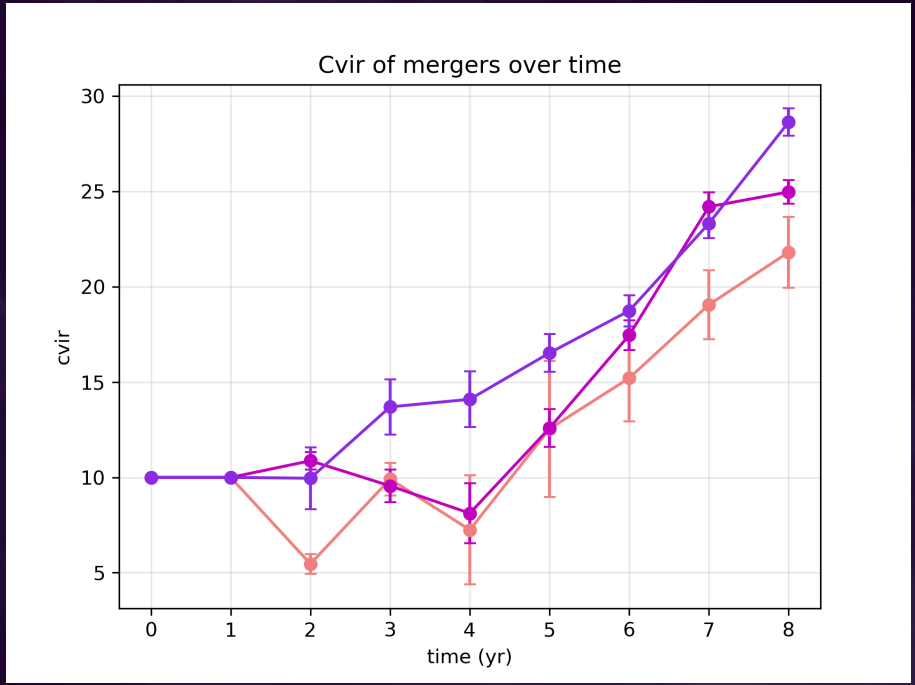
Cvir of minor merger over time
cvir
time (yr)

Virialization and total energy checks



Concentration through time

Major and minor trends are nearly parallel, but Mid trend jumps between both at different times



Conclusions

Our results reflect the behavior of halos in Wang et al. 2020's FIRE-2 simulation study

-> cvir decreases following merge then increases indefinitely

Why this matters so much:

-> small satellite halos far outnumber big halos with galaxies

-> they have a larger effect on the internal density than expected

Future endeavors for this project

Sources

1. Springel V n.d. *Image from National Geographic* (Image) National Geographic. Available at: URL (Accessed: 8 December 2025)
2. Powell M n.d. *Image from Stanford.edu* (Image) Stanford University. Available at: URL (Accessed: 8 December 2025)
3. K. Wang, Y.-Y. Mao, A. R. Zentner, J. U. Lange, F. C. van den Bosch, and R. H. Wechsler, Concentrations of dark haloes emerge from their merger histories, Mon Not R Astron Soc 498, 4450 (2020).
4. J. F. Navarro, C. S. Frenk, and S. D. M. White, The Structure of Cold Dark Matter Halos, The Astrophysical Journal 462, 563 (1996).
5. A. A. Klypin, S. Trujillo-Gomez, and J. Primack, Dark Matter Halos in the Standard Cosmological Model: Results from the Bolshoi Simulation, The Astrophysical Journal 740, 102 (2011).